校园实验室设备预约与借用管理系统

完整实体关系图谱

40个实体 / 341个字段 / 71条物理外键 / PostgreSQL

🔑 主键 › R01 ⟞ Parent.id 外键与目标实体 ? 可空 · UQ 单列唯一

实线：域内及主业务外键。共享责任主体引用以字段目标完整展开。
可隐藏跟支件可切换“共享引用”图层，检查全部71条连接。

A 身份与权限

共享身份、岗位和管理范围

6 实体

User A01
身份映射
core_user

🔑 id	bigint	
username	varchar(150)	UQ
school_no	varchar(32)	UQ
password	varchar(128)	
last_login	timestamptz	?
is_superuser	boolean	
first_name	varchar(150)	
last_name	varchar(150)	
email	varchar(254)	
is_staff	boolean	
is_active	boolean	
date_joined	timestamptz	
display_name	varchar(40)	

11 fields 主键 / 外键 / 约束字段

Role A02
业务角色
core_role

🔑 id	bigint	
code	varchar(20)	UQ
name	varchar(40)	

3 fields 主键 / 外键 / 约束字段

LabGrant A04
实验室管理范围
core_labgrant

🔑 id	bigint	
› user_id R05 ⟞ User.id	bigint	AUI
› lab_id R06 ⟞ Lab.id	bigint	AUI
granted_at	timestamptz	

4 fields 主键 / 外键 / 约束字段

UserRole A03
用户角色关联
core_userrole

🔑 id	bigint	
› user_id R05 ⟞ User.id	bigint	AUI
› role_id R02 ⟞ Role.id	bigint	AUI
› granted_by_id R05 ⟞ User.id	bigint	? AUI
granted_at	timestamptz	

5 fields 主键 / 外键 / 约束字段

QualificationType A05
培训资格类型
core_qualificationtype

🔑 id	bigint	
code	varchar(32)	UQ
name	varchar(80)	

3 fields 主键 / 外键 / 约束字段

UserQualification A06
用户培训资格
core_userqualification

🔑 id	bigint	
› user_id R07 ⟞ User.id	bigint	AUI
› qualification_id R08 ⟞ QualificationType.id	bigint	AUI
› granted_by_id R05 ⟞ User.id	bigint	? AUI
issued_at	timestamptz	
expires_at	timestamptz	?
revoked	boolean	

7 fields 主键 / 外键 / 约束字段

B 资源与开放

校区、楼宇、资源主档和开放日历

9 实体

Campus B01
校区
core_campus

🔑 id	bigint	
code	varchar(32)	UQ
name	varchar(80)	
is_active	boolean	

4 fields 主键 / 外键 / 约束字段

Building B02
楼宇
core_building

🔑 id	bigint	
› campus_id R04 ⟞ Campus.id	bigint	AUI
code	varchar(32)	
name	varchar(80)	
address_label	varchar(160)	
is_active	boolean	

6 fields 主键 / 外键 / 约束字段

Lab B03
实验室
core_lab

🔑 id	bigint	
code	varchar(32)	UQ
name	varchar(80)	
location	varchar(120)	
description	text	
is_active	boolean	

6 fields 主键 / 外键 / 约束字段

OpeningPeriod B09
日间开放时段
core_openingperiod

🔑 id	bigint	
› lab_id R06 ⟞ Lab.id	bigint	AUI
› resource_id R05 ⟞ Resource.id	bigint	? AUI
weekday	smallint	
starts_at	time	
ends_at	time	
closed_all_day	boolean	
valid_from	date	
valid_to	date	?

9 fields 主键 / 外键 / 约束字段

Accessory B07
设备选件和配置
core_accessory

🔑 id	bigint	
› resource_id R13 ⟞ Resource.id	bigint	AUI
code	varchar(32)	
name	varchar(80)	
quantity	integer	
is_active	boolean	

6 fields 主键 / 外键 / 约束字段

CalendarException B09
单日开放例外版本
core_calendarexception

🔑 id	bigint	
› lab_id lab ⟞ Lab.id	bigint	AUI
› actor_id R08 ⟞ User.id	bigint	AUI
day	date	
closed	boolean	
starts_at	time	?
ends_at	time	?
active	boolean	
reason	text	
created_at	timestamptz	

10 fields 主键 / 外键 / 约束字段

Resource B04
可预约资源
core_resource

🔑 id	bigint	
› lab_id R15 ⟞ Lab.id	bigint	AUI
code	varchar(40)	UQ
name	varchar(80)	
kind	varchar(16)	
description	text	
service_status	varchar(20)	
manual_hold	varchar(16)	
created_at	timestamptz	
updated_at	timestamptz	

10 fields 主键 / 外键 / 约束字段

EquipmentDetail B05
设备属性
core_equipmentdetail

🔑 id	bigint	
› resource_id R11 ⟞ Resource.id	bigint	UQ AUI
› qualification_id R12 ⟞ QualificationType.id	bigint	? AUI
› access_room_id R13 ⟞ Resource.id	bigint	? AUI
category	varchar(16)	
risk	varchar(16)	
physical_status	varchar(16)	
portable	boolean	
handover_location	varchar(120)	

9 fields 主键 / 外键 / 约束字段

RoomDetail B06
场地属性
core_roomdetail

🔑 id	bigint	
› resource_id R14 ⟞ Resource.id	bigint	UQ AUI
› building_id R15 ⟞ Building.id	bigint	? AUI
floor_label	varchar(40)	
room_number	varchar(40)	
location_label	varchar(160)	
capacity	integer	
space_category	varchar(30)	
facilities	text	
safety_note	text	

10 fields 主键 / 外键 / 约束字段

C 预约与规则

申请、最终决定和规则版本

6 实体

Reservation C01
资源预约
core_reservation

🔑 id	bigint	
› applicant_id R40 ⟞ User.id	bigint	AUI
› resource_id R20 ⟞ Resource.id	bigint	AUI
code	varchar(40)	UQ
starts_at	timestamptz	
ends_at	timestamptz	
purpose	text	
participants	integer	
status	varchar(16)	
approval_mode	varchar(16)	
request_key	uuid	
request_fingerprint	varchar(64)	
policy_snapshot	jsonb	
reason	text	
version	integer	
review_required	boolean	
review_reason	text	
arrived_at	timestamptz	?
late_start_until	timestamptz	?
created_at	timestamptz	

20 fields 主键 / 外键 / 约束字段

ApprovalDecision C02
自动或人工确认决定
core_approvaldecision

🔑 id	bigint	
› reservation_id R22 ⟞ Reservation.id	bigint	UQ C01
› actor_id R23 ⟞ User.id	bigint	? AUI
mode	varchar(16)	
approved	boolean	
reason	text	
decided_at	timestamptz	

7 fields 主键 / 外键 / 约束字段

ReservationReview C05
预约复核历史
core_reservationreview

🔑 id	bigint	
› reservation_id R24 ⟞ Reservation.id	bigint	C01
› actor_id R25 ⟞ User.id	bigint	AUI
accepted	boolean	
reason	text	
snapshot	jsonb	
created_at	timestamptz	

7 fields 主键 / 外键 / 约束字段

Policy C04
当前业务参数
core_policy

🔑 id	bigint
version	integer
window_days	integer
slot_minutes	integer
max_hours	integer
equipment_quota	integer
room_quota	integer
simultaneous_equipment	integer
grace_minutes	integer
must_delivery_failure	boolean
updated_at	timestamptz

11 fields 主键 / 外键 / 约束字段

PolicyRevision C06
参数历史版本
core_policyrevision

🔑 id	bigint	
› policy_id R26 ⟞ Policy.id	bigint	C04
› actor_id R27 ⟞ User.id	bigint	? AUI
version	integer	
data	jsonb	
reason	text	
created_at	timestamptz	

7 fields 主键 / 外键 / 约束字段

ReviewNote C03
审核补充记录
core_reviewnote

🔑 id	bigint	
› reservation_id R42 ⟞ Reservation.id	bigint	C01
› actor_id R43 ⟞ User.id	bigint	AUI
kind	varchar(16)	
content	text	
evidence_ref	varchar(240)	
created_at	timestamptz	

7 fields 主键 / 外键 / 约束字段

D 现场使用

设备交接与场地实际使用

5 实体

Loan D01
设备借出事实
core_loan

🔑 id	bigint	
› reservation_id R34 ⟞ Reservation.id	bigint	UQ C01
› operator_id R35 ⟞ User.id	bigint	AUI
code	varchar(40)	UQ
arrived_at	timestamptz	
service_started_at	timestamptz	
handed_over_at	timestamptz	
due_at	timestamptz	
tag_snapshot	varchar(40)	
condition_out	text	

10 fields 主键 / 外键 / 约束字段

ReturnRecord D02
设备归还事实
core_returnrecord

🔑 id	bigint	
› loan_id R36 ⟞ Loan.id	bigint	UQ D01
› operator_id R37 ⟞ User.id	bigint	AUI
code	varchar(40)	UQ
returned_at	timestamptz	
damaged	boolean	
missing	boolean	
fault	boolean	
note	text	

9 fields 主键 / 外键 / 约束字段

RoomUse D04
场地实际使用与检查
core_roomuse

🔑 id	bigint	
› reservation_id R40 ⟞ Reservation.id	bigint	UQ C01
› started_by_id R41 ⟞ User.id	bigint	AUI
› ended_by_id R42 ⟞ User.id	bigint	? AUI
arrived_at	timestamptz	
started_at	timestamptz	
actual_participants	integer	
start_note	text	
ended_at	timestamptz	?
end_note	text	
abnormal	boolean	

11 fields 主键 / 外键 / 约束字段

HandoverItem D03
配件交接快照
core_handoveritem

🔑 id	bigint	
› loan_id R38 ⟞ Loan.id	bigint	D01
› accessory_id R39 ⟞ Accessory.id	bigint	? AUI
code_snapshot	varchar(32)	
name_snapshot	varchar(80)	
expected_qty	integer	
out_qty	integer	
out_condition	varchar(160)	
return_qty	integer	?
return_condition	varchar(160)	

10 fields 主键 / 外键 / 约束字段

Coordination D06
预约协调凭据
core_coordination

🔑 id	bigint	
› reservation_id R56 ⟞ Reservation.id	bigint	C01
› actor_id R57 ⟞ User.id	bigint	AUI
action	varchar(16)	
reason	text	
old_state	jsonb	
new_state	jsonb	
created_at	timestamptz	

9 fields 主键 / 外键 / 约束字段

E 异常与维修

证据、工单和放立验收

5 实体

Incident E01
资源异常事实
core_incident

🔑 id	bigint	
› resource_id R46 ⟞ Resource.id	bigint	E03
› reporter_id R47 ⟞ User.id	bigint	AUI
› reservation_id R48 ⟞ Reservation.id	bigint	? C01
› confirmed_by_id R49 ⟞ User.id	bigint	? AUI
code	varchar(40)	UQ
kind	varchar(16)	
description	text	
blocks_service	boolean	
occurred_at	timestamptz	
reported_at	timestamptz	
status	varchar(16)	
conclusion	text	
resolved_at	timestamptz	?

14 fields 主键 / 外键 / 约束字段

Evidence E02
异常证据索引
core_evidence

🔑 id	bigint	
› incident_id R50 ⟞ Incident.id	bigint	AUI
› uploaded_by_id R40 ⟞ User.id	bigint	AUI
file	varchar(200)	
original_name	varchar(160)	
sha256	varchar(64)	
uploaded_at	timestamptz	

7 fields 主键 / 外键 / 约束字段

IncidentResponse E08
异常确认与异议记录
core_incidentresponse

🔑 id	bigint	
› incident_id R52 ⟞ Incident.id	bigint	AUI
› actor_id R53 ⟞ User.id	bigint	AUI
kind	varchar(16)	
content	text	
created_at	timestamptz	

6 fields 主键 / 外键 / 约束字段

WorkOrder E03
维修工单
core_workorder

🔑 id	bigint	
› incident_id R45 ⟞ Incident.id	bigint	E01
› created_by_id R30 ⟞ User.id	bigint	AUI
› assigned_to_id R31 ⟞ User.id	bigint	AUI
› accepted_by_id R32 ⟞ User.id	bigint	? AUI
code	varchar(40)	UQ
status	varchar(16)	
priority	varchar(12)	
result	text	
finished_at	timestamptz	?
accepted_at	timestamptz	?
created_at	timestamptz	

12 fields 主键 / 外键 / 约束字段

WorkOrderEvent E04
维修及验收过程
core_workorderevent

🔑 id	bigint	
› work_order_id R33 ⟞ WorkOrder.id	bigint	E03
› actor_id R54 ⟞ User.id	bigint	AUI
old_status	varchar(16)	
new_status	varchar(16)	
note	text	
created_at	timestamptz	

7 fields 主键 / 外键 / 约束字段

F 审计与通知

状态区间和可追踪投递

7 实体

AuditEvent F01
不可覆盖审计事件
core_auditevent

🔑 id	bigint	
› actor_id R15 ⟞ User.id	bigint	? AUI
› lab_id R16 ⟞ Lab.id	bigint	? BUI
dedupe_key	varchar(160)	? UQ
actor_kind	varchar(8)	
entity_type	varchar(40)	
entity_id	bigint	
action	varchar(40)	
old_state	jsonb	
new_state	jsonb	
reason	text	
request_id	uuid	
created_at	timestamptz	

14 fields 主键 / 外键 / 约束字段

ServicePeriod F02
资源服务状态区间
core_serviceperiod

🔑 id	bigint	
› resource_id R17 ⟞ Resource.id	bigint	BUI
› audit_id R18 ⟞ AuditEvent.id	bigint	UQ F01
status	varchar(20)	
starts_at	timestamptz	
ends_at	timestamptz	?

6 fields 主键 / 外键 / 约束字段

MockReceipt F06
模拟投递去重凭据
core_mockreceipt

🔑 id	bigint	
› delivery_id R63 ⟞ Delivery.id	bigint	UQ F04
received_at	timestamptz	

3 fields 主键 / 外键 / 约束字段

Notification F03
接收人通知
core_notification

🔑 id	bigint	
› audit_id R62 ⟞ AuditEvent.id	bigint	F01
› recipient_id R60 ⟞ User.id	bigint	AUI
code	varchar(40)	UQ
kind	varchar(32)	
content	text	
created_at	timestamptz	
read_at	timestamptz	?

9 fields 主键 / 外键 / 约束字段

Delivery F04
渠道投递任务
core_delivery

🔑 id	bigint	
› notification_id R61 ⟞ Notification.id	bigint	F03
channel	varchar(12)	
status	varchar(16)	
retry_count	smallint	
next_attempt_at	timestamptz	
locked_until	timestamptz	?
last_error	text	

9 fields 主键 / 外键 / 约束字段

DeliveryAttempt F05
投递尝试
core_deliveryattempt

🔑 id	bigint	
› delivery_id R62 ⟞ Delivery.id	bigint	F04
attempt_no	smallint	
outcome	varchar(12)	
detail	text	
attempted_at	timestamptz	

6 fields 主键 / 外键 / 约束字段

JobHeartbeat F07
后台任务健康记录
core_jobheartbeat

🔑 id	bigint	
name	varchar(40)	UQ
last_success	timestamptz	?
last_error_at	timestamptz	?
error_code	varchar(20)	

5 fields 主键 / 外键 / 约束字段

G 资源安全信息

适用于管理单元或具体资源。具单源与复核版本的安全信息

2 实体

SafetyProcedure G01
安全信息版本
core_safetyprocedure

🔑 id	bigint	
› lab_scope_id R58 ⟞ Lab.id	bigint	? BUI
› resource_id R57 ⟞ Resource.id	bigint	? BUI
› created_by_id R59 ⟞ User.id	bigint	AUI
› reviewed_by_id R60 ⟞ User.id	bigint	? AUI
› published_by_id R61 ⟞ User.id	bigint	? AUI
version	integer	
title	varchar(160)	
risk_categories	text	
applicable_users	text	
prerequisites	text	
required_ppe	text	
pre_use_checks	text	
operating_cautions	text	
prohibited_actions	text	
incident_route	text	
source_title	varchar(200)	
source_reference	varchar(500)	
source_version	varchar(80)	
source_effective_date	date	?
effective_from	date	?
review_due_date	date	?
requires_acknowledgement	boolean	
status	varchar(20)	
submitted_at	timestamptz	?
reviewed_at	timestamptz	?
review_note	text	
published_at	timestamptz	?
change_reason	text	
created_at	timestamptz	

30 fields 主键 / 外键 / 约束字段

SafetyAcknowledgement G02
安全信息阅读确认
core_safetyacknowledgement

🔑 id	bigint	
› procedure_id R23 ⟞ SafetyProcedure.id	bigint	G01
› user_id R22 ⟞ User.id	bigint	AUI
acknowledged_at	timestamptz	

4 fields 主键 / 外键 / 约束字段

阅读顺序：资源 → 预约 → 现场使用 → 异常维修；身份、规则与审计为共用支撑。

实体代码用于跨域定位：R01–R71对应完整关系表引。实体、字段及外键由实际模型导出。

关键约束

待审核不占资源；确认与使用中的重叠区间互斥。
设备与场地分别记录实际使用，审计逻辑引用不负责为仲裁。

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